TEAM 37
ZONE: 2

Cardinal

Reinforcement Learning-Based LLM Query Optimization

Aiden Lee

Anirudh Bharadwaj

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Penn Engineering
University of Pennsylvania

Faculty Mentor: Ryan Marcus; TA Mentor: Kevin Dong

MOTIVATION

Inefficient SQL causes slower run times and increased operating costs; we need a more efficient and reliable query optimization solution.

MAIN FEATURES

Execution-Grounded Dataset

- Built a dataset of **18k+ executed SQL queries**, capturing query text, execution plans, and real runtime metrics from PostgreSQL
- **Autoregressive Plan Generation**
  Language models generate **PostgreSQL execution plans (join order, indexes, operators)** directly from SQL queries

Two-Stage Model Training

1. **Supervised Fine-Tuning (SFT)** on execution-grounded data
2. **GRPO Reinforcement Learning** using real query execution metrics

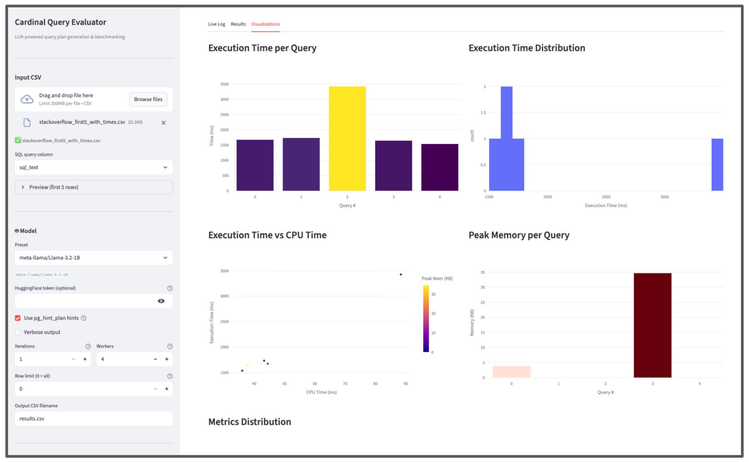
Unified Evaluation Framework

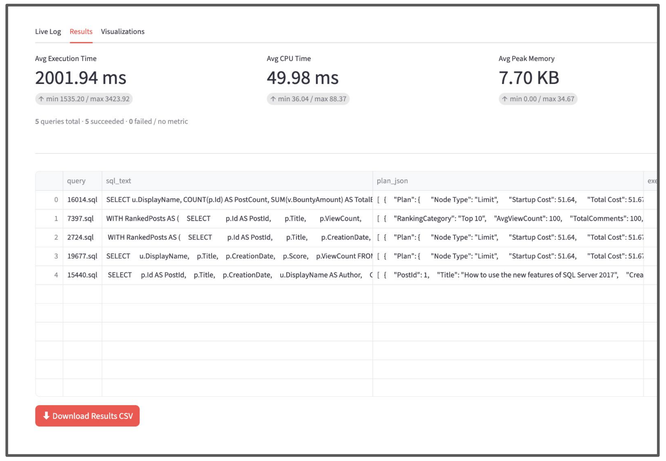
- Benchmark models using execution time, CPU cycles, and peak memory across a shared testing pipeline

SOCIAL IMPACT

Productivity: Enhances execution speed of query wherever database is used
Efficiency: Enhances the cost and energy of running queries

PROJECT IN ACTION





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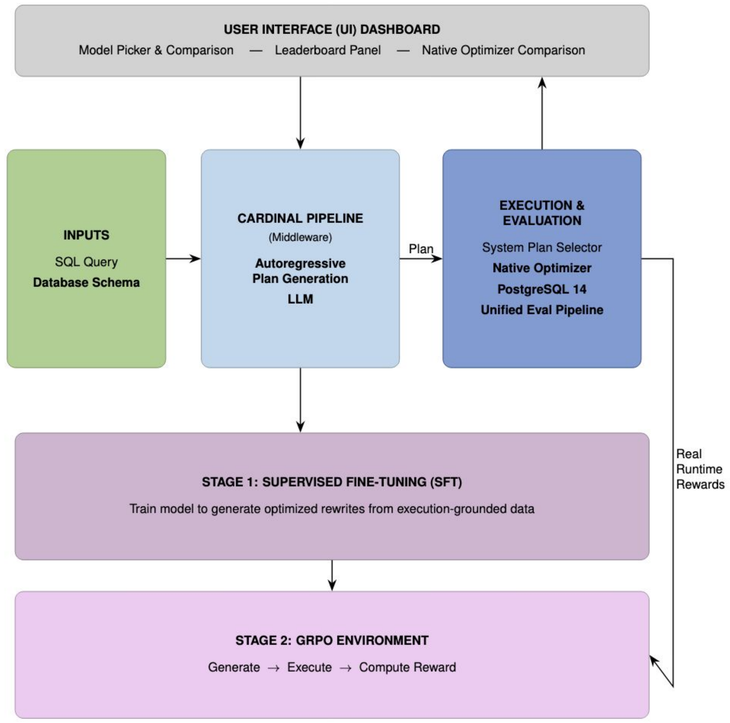

TECH STACK

LLaMA by Meta

Streamlit

FUTURE WORK

Scaling Model: Train a larger model for better performance on actual production environments

Platform Deployment: Deploy the UI, provide full pipeline for RL-finetuning custom models, and allow metric tracking

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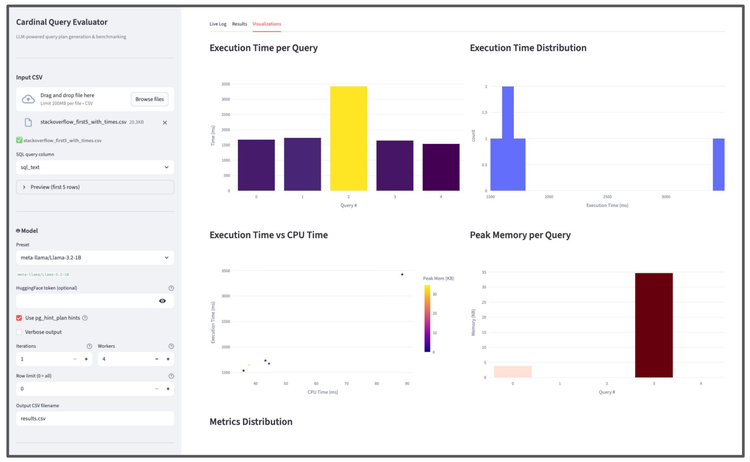

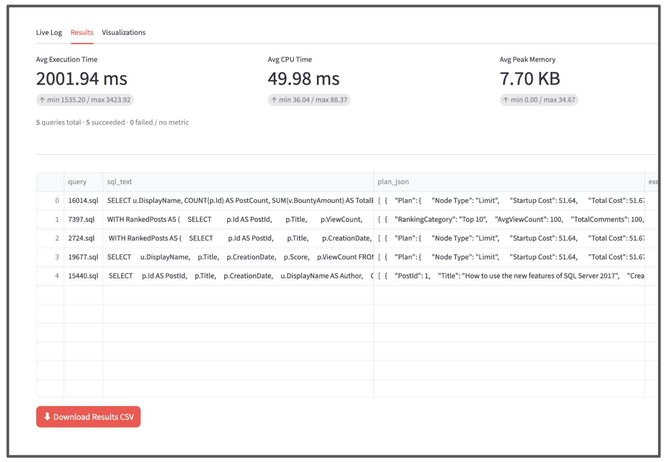

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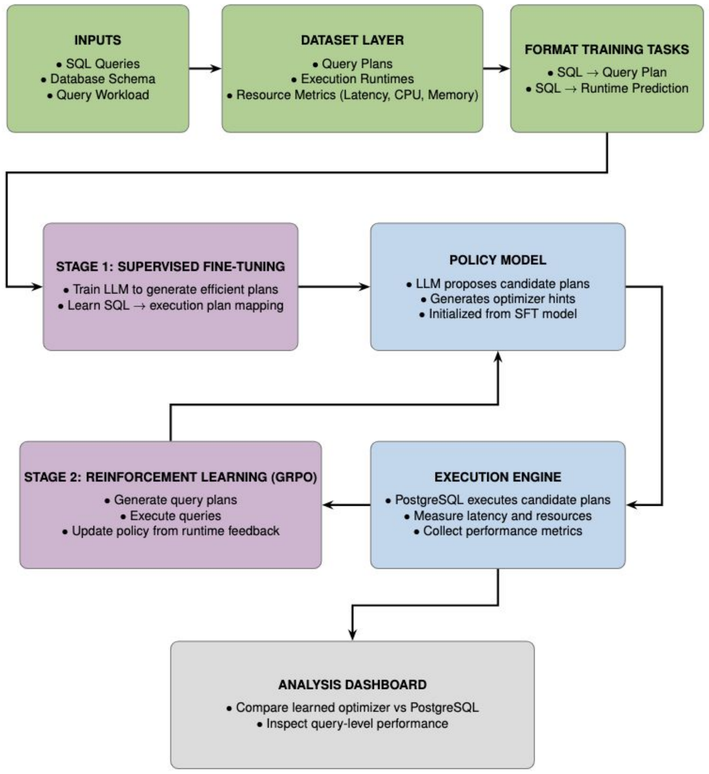

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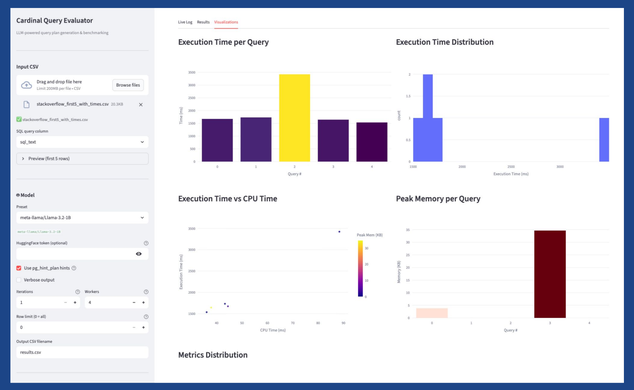

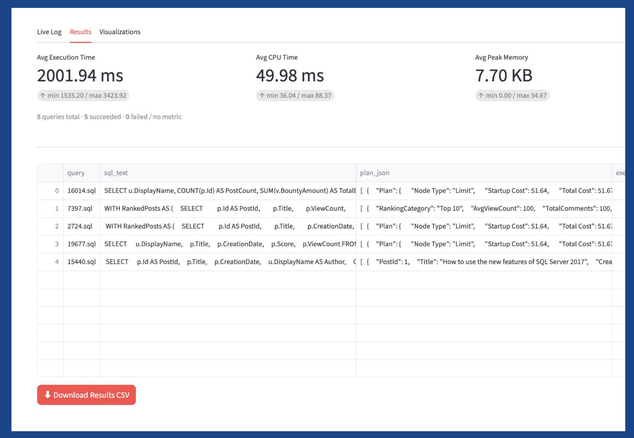

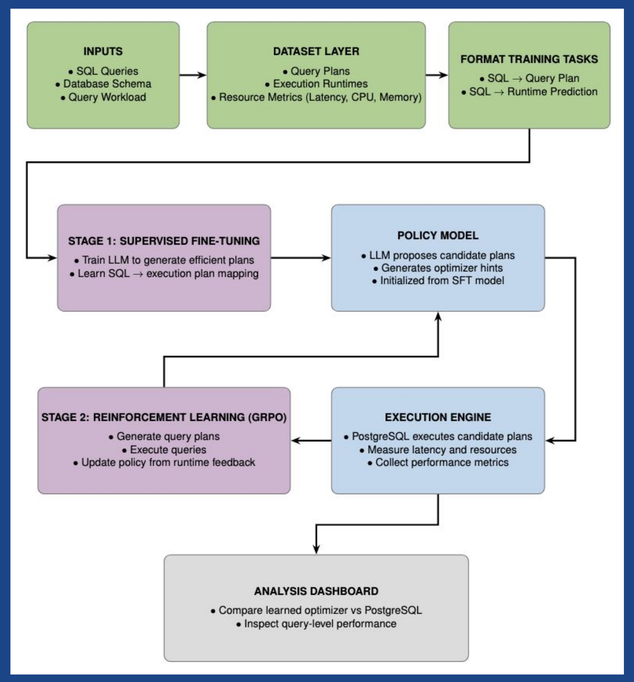

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